NUCLEAR POWER CORPORATION OF INDIA LIMITED

WELDING PROCEDURE SPECIFICATION (WPS) - ASME QW-482

SECTION 1 - IDENTIFICATION			
WPS NO.	NPCIL-GT-88-ATT	REVISION NO.	R-1
DATE	2014-07-07	QUALIFICATION STATUS	To be qualified
SUPPORTING PQR NO.(S)	To be made by user organization		

QA APPROVED

SECTION 2 - WELDING PROCESSES			
WELDING PROCESS (ROOT)	GTAW	WELDING PROCESS (FILL)	GTAW
TYPE	Manual	JOINT DESIGN	Groove
BACKING	With or Without	BACKING MATERIAL	As per design document

SECTION 4 - BASE METALS (QW-403)			
P-NO.	P-No. 8 Tube	TO P-NO.	P-No. 8 Tube
MATERIAL SPEC / GRADE	Synthetic but code-style material family		
THICKNESS RANGE (GROOVE)	1.2 mm to 1.5 mm	THICKNESS RANGE (FILLET)	All Thickness

SECTION 5 - POSITIONS (QW-405)			
POSITION(S) OF GROOVE	All	WELDING PROGRESSION	Vertical Uphill
POSITION(S) OF FILLET	All	PWHT	Not Required

PREPARED BY

REVIEWED BY (QA)

APPROVED BY

SECTION 6 - PREHEAT & PWHT (QW-406 / QW-407)			
PREHEAT TEMP, MIN	10.0 °C	INTERPASS TEMP, MAX	150.0
PWHT REQUIREMENTS	Not Required		

SECTION 8 - ELECTRICAL CHARACTERISTICS (QW-409)	
CURRENT / POLARITY	DC-EN
TRAVEL SPEED RANGE	40-80 mm/min

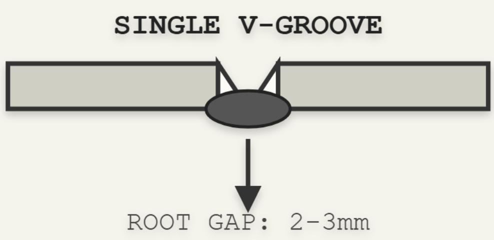
SECTION 9 - TECHNIQUE (QW-410)	
STRING OR WEAVE BEAD	Root: Stringer / Fill: Weave
CLEANING METHOD	Brushing or Grinding
PEENING ALLOWED	False

SECTION 10 - FILLER METALS (QW-404)			
SFA SPECIFICATION	NA	AWS CLASSIFICATION	Autogenous
F-NO.	varies	A-NO.	varies

NOTES & RESTRICTIONS:

Synthetic WPS record used for contrast and edge cases.

JOINT SCHEMATIC



WPS NO. : NPCIL-GT-88-ATT

SECTION 6 - WELDING PARAMETERS & CONSUMABLES

Pass	Process	Filler Spec	Filler Class	Size (mm)	Amps	Volts	Speed (mm/min)
Root	As Spec	SFA 5.1/5.18	As Spec	2.4/2.5	60-140	18-30	40-80
Fill 1	As Spec	SFA 5.1/5.18	As Spec	3.15	80-160	20-30	40-80
Fill 2	As Spec	SFA 5.1/5.18	As Spec	3.15	90-180	22-32	40-80
Cap	As Spec	SFA 5.1/5.18	As Spec	3.15/4.0	90-190	22-32	40-80

SECTION 7 - TECHNIQUE

Stringer or Weave Bead:	Stringer for Root, Weave for Fill/Cap
Orifice or Gas Cup Size:	10mm to 16mm (for GTAW)
Initial and Interpass Cleaning:	Brushing, Grinding, or Chipping
Method of Back Gouging:	Grinding or Machining (if applicable)
Oscillation:	Max 3 times electrode diameter
Multiple or Single Pass per Side:	Multiple
Multiple or Single Electrodes:	Single
Peening:	Not Allowed